Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Full-Stack · PropTech · Financial Tooling · AWS · AI/LLM Integration · Production SaaS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia — DC area remote

PROFESSIONAL SUMMARY

I built the financial reporting layer property managers and owners use to run their portfolios — 8 years at an AppFolio competitor. At Entrata: owned **owner-facing financial workflows end-to-end** — distribution reporting, financial statement generation, account reconciliation, and owner portal financials — across 20+ production modules (1M+ records, 100K+ MAU, 99.5% uptime, 120+ Agile sprints). The Accounting team builds the financial foundation those workflows sit on; I built the reporting layer on top, which means I know exactly what accounting data needs to look like downstream. The JD lists AI/LLM integration as nice-to-have — I built open-source multi-agent evaluation infrastructure with **LLM-as-judge, embeddings, and context-aware retrieval** in production: AgentProbe (live on GitHub). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built end-to-end ML pipelines (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest anomaly detection across 3 research pipelines; automated statistical reporting with Python + Quarto driving **40% improvement** in stakeholder comprehension.

Entrata
Jan 2021 – Aug 2024

Senior Software Engineer — Financial Tooling · Owner Portal · Full-Stack · AWS · Production AI

- Owned **owner-facing financial workflows** end-to-end — distribution reporting, financial statement generation, **account type expansion**, account reconciliation, and owner portal financials across **20+ modules** (1M+ records, 100K+ MAU, 99.5% uptime, 120+ Agile sprints); optimized **200+ SQL queries** (30% performance gain) across all client environments.
- Led three platform-level initiatives: Docker microservices rollout (**50% speedup**), AWS S3/RDS/Glue migration (**100% PHI compliance**, zero incidents), and Entrata's **first production AI pipeline on 1.6M real financial records** (PyTorch, OpenAI API) with quality gates and HITL review.
- Mentored 5+ engineers, **300+ code reviews**; built LIHTC (Tax Credit) and HUD Section 8/HCV compliance modules with deterministic rules engines — **80% manual workload eliminated**, full audit trails, zero compliance violations across 100K+ units.

Entrata
Jan 2019 – Dec 2020

Software Engineer — Compliance Automation · API Integrations · Data Pipelines

- Built compliance automation processing **millions of records** — income limits, household composition, asset verification thresholds in deterministic systems with full audit trails; built **API integrations and webhooks** across property management systems; conducted 300+ peer code reviews; mentored 5+ junior engineers.

Entrata
Jan 2017 – Dec 2018

Associate Engineer — Full-Stack · Python · PostgreSQL · REST APIs

- Built and shipped full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management and owner-facing financial workflows; 0→1 prototyping through production rollout across 100K+ unit portfolios.

PROJECTS

AgentProbe: Multi-Agent AI Eval Framework
`MULTI-AGENT` `LLM-AS-JUDGE`

Python · LangGraph · FastAPI · ChromaDB · Embeddings · OpenAI · Anthropic · Docker · GitHub Actions · React — github.com/venkatalakshmiparimala/AgentProbe

- 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) — direct coverage of JD nice-to-haves: **Agents, Embeddings, Context Windows, LLM integration**; 5-metric LLM-as-judge eval engine with CI/CD regression gates blocking failing runs before production; ChromaDB RAG pipeline; React observability dashboard.

Lease: Agentic Compliance Pipeline for Legal Documents
`PROPTECH` `AGENTIC RAG`

Python · LangGraph · ChromaDB · Deterministic Rules Engine · LLM-as-Judge · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **93.6% precision, 87.9% F1** on 1,240 residential lease PDFs — agentic LangGraph RAG pipeline with extraction, embedding, retrieval, and deterministic rules engine for fully auditable compliance outputs; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge and CI/CD regression gates.

AI on FHIR: HIPAA-Compliant Medical Records NLP Platform
`HEALTHCARE`

Python · Flask · React · FHIR APIs · BERT · LLMs · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into FHIR API calls via LLMs + BERT + embeddings; HIPAA-compliant RBAC, end-to-end from design to deployment.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

ML systems, statistical modeling, data engineering, AI pipelines, system design, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, PHP, R; FastAPI, Django, Flask, REST APIs, webhooks, API integrations, RabbitMQ, event-driven distributed systems; Ruby on Rails (fast ramp — OOP/MVC foundation in Python/PHP)

AI & Systems: LangGraph, LangChain, Agentic AI, Multi-Agent Orchestration, Tool Calling, LLM Integration, LLM-as-Judge, LLM Evaluation, HITL Feedback Loops, RAG Pipelines, Embeddings, Vector Stores (ChromaDB, pgvector), Context Windows, OpenAI API, Anthropic API, PyTorch, BERT, NLP, system design, scalable architecture, design patterns

Infrastructure & Cloud: AWS (S3, EC2, RDS, Glue), Docker, Microservices, SOA, CI/CD, GitHub Actions, React, Redux, PostgreSQL, MySQL, HIPAA, PHI compliance, test-driven development (TDD), Agile/Scrum, pair programming, Git, Linux